Auto Scaling Groups.

1. What is ASG?

An **Auto Scaling Group (ASG)** in Amazon Web Services automatically manages a group of Amazon EC2 instances.

It ensures:

- The **right number of instances**
- At the **right time**
- Based on **demand**

2. Why ASG is Used

Without ASG:

- Manual scaling → slow + error-prone
- Over-provision → waste money
- Under-provision → system crash

ASG solves all three:

- High availability
- Cost optimization
- Fault tolerance

To create an ASG we have to create on Launch Template using the template the EC2 Creation will be easier. And ASG uses the Template to Maintain the Desired State.

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - *required*

my-template

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '*', '@'.

Template version description

A prod webserver for MyApp

Max 255 chars

Auto Scaling guidance | Info

Select this if you intend to use this template with EC2 Auto Scaling

☑ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

▶ **Template tags**

▶ **Source template**

Launch template contents

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

▼ Application and OS Images (Amazon Machine Image) - required Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Search our full catalog including 1000s of application and OS images

Recents | Quick Start

Amazon Linux | macOS | Ubuntu | Windows | Red Hat | SUSE Linux | Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI — Free tier eligible

ami-0931307dcdc2a28c9 (64-bit (x86), uefi-preferred) / ami-0e6f3e69477999e5d (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

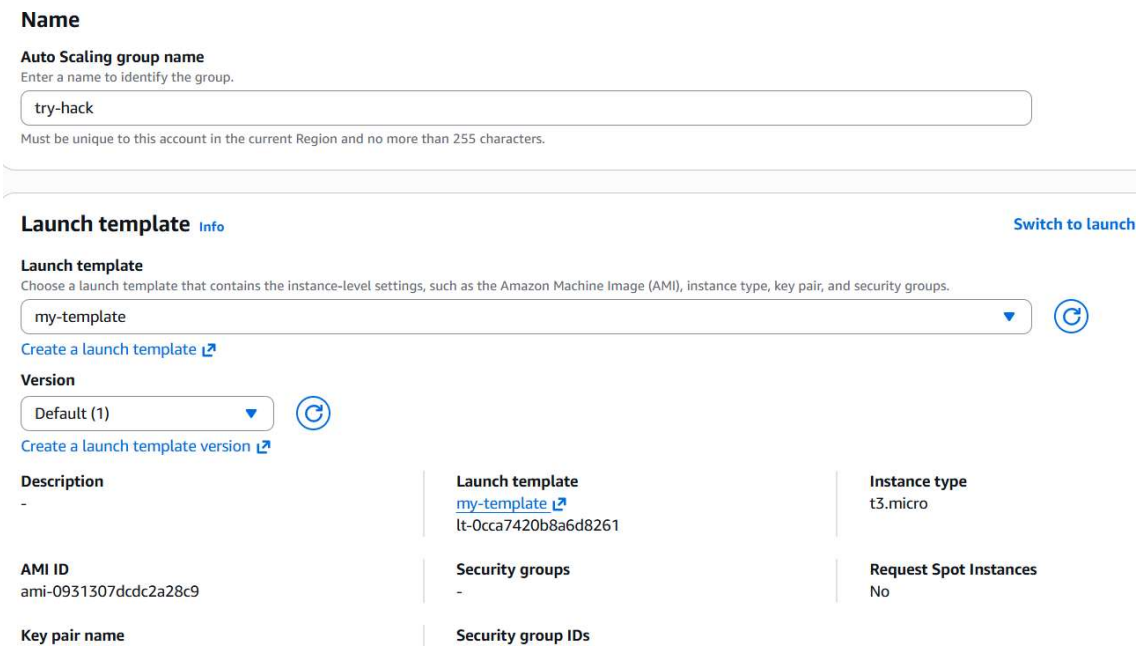
Launch Templates (1) Info

Actions | Create launch template

Search

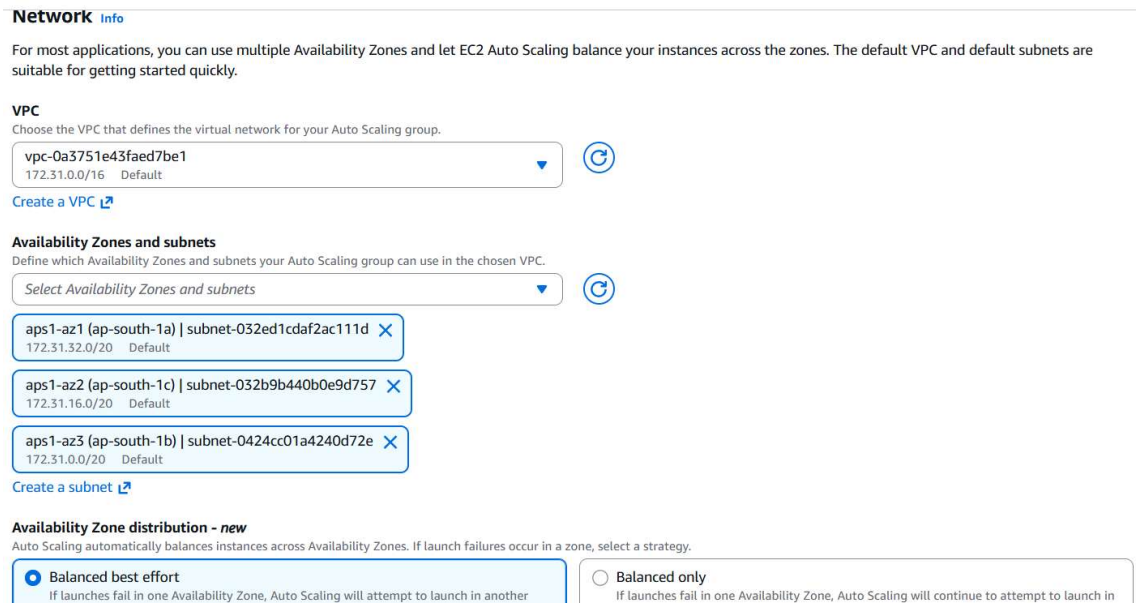
Launch Template ID	Launch Template Name	Default Version	Latest Version	Create Time	Created By
lt-0cca7420b8a6d8261	my-template	1	1	2026-03-28T10:40:41.000Z	arn:aws:iam::089665

Then I want to create the ASG Using this Template.



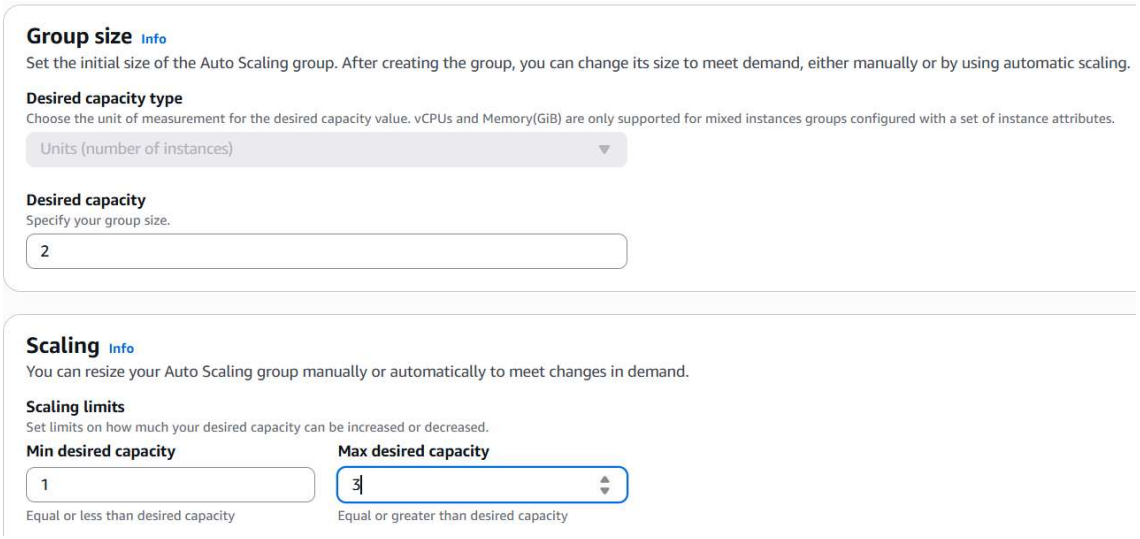
After selecting the template we will see the Selected Instance types.

Then we have to configure network settings.

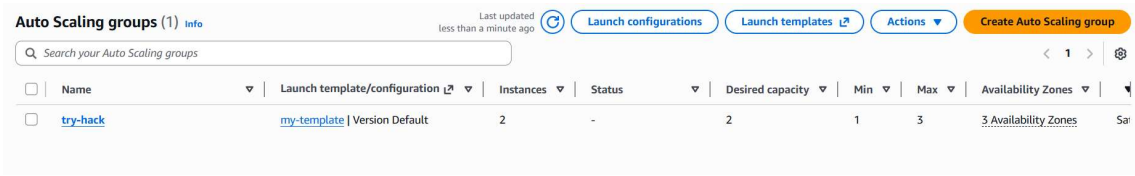


After that we have to select the with Load Balancer or without Load balancer.

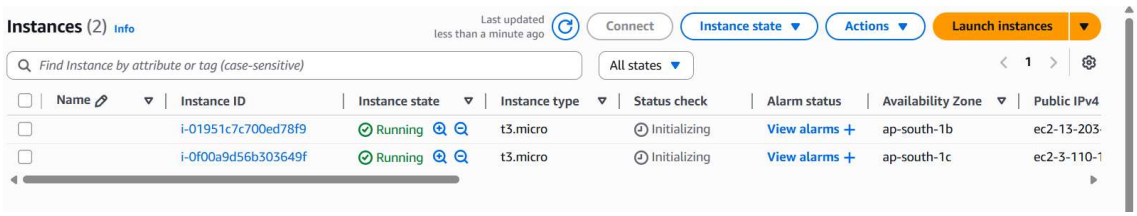
After that we have to configure the desired Instance Running State.



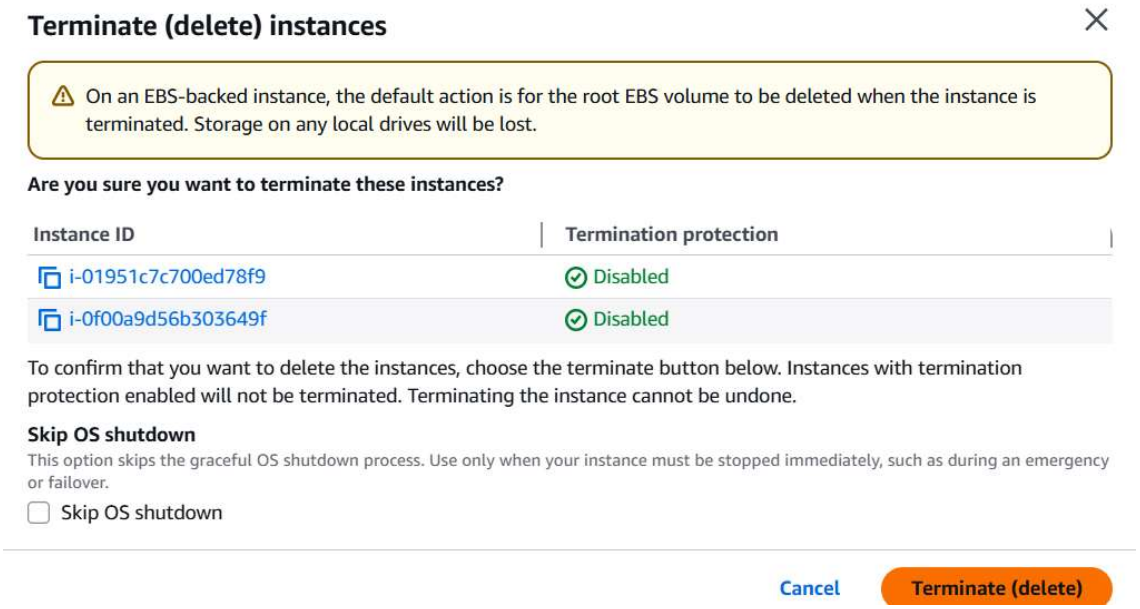
Created ASG.



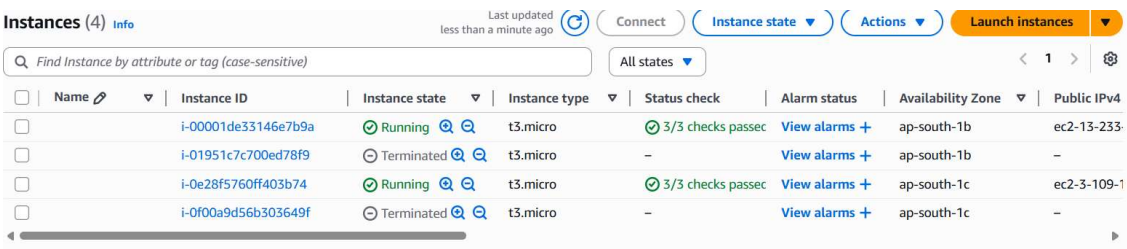
After the ASG Creation the instances are automatically started creating using the ASG with the help of Launch Template.



When I delete the instance.

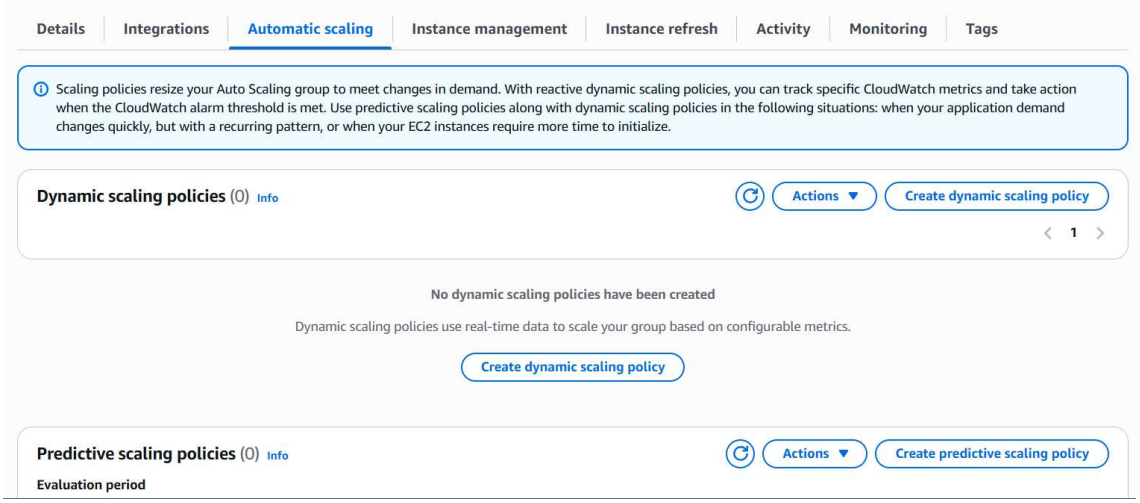


It will recreate itself new Two Instances.



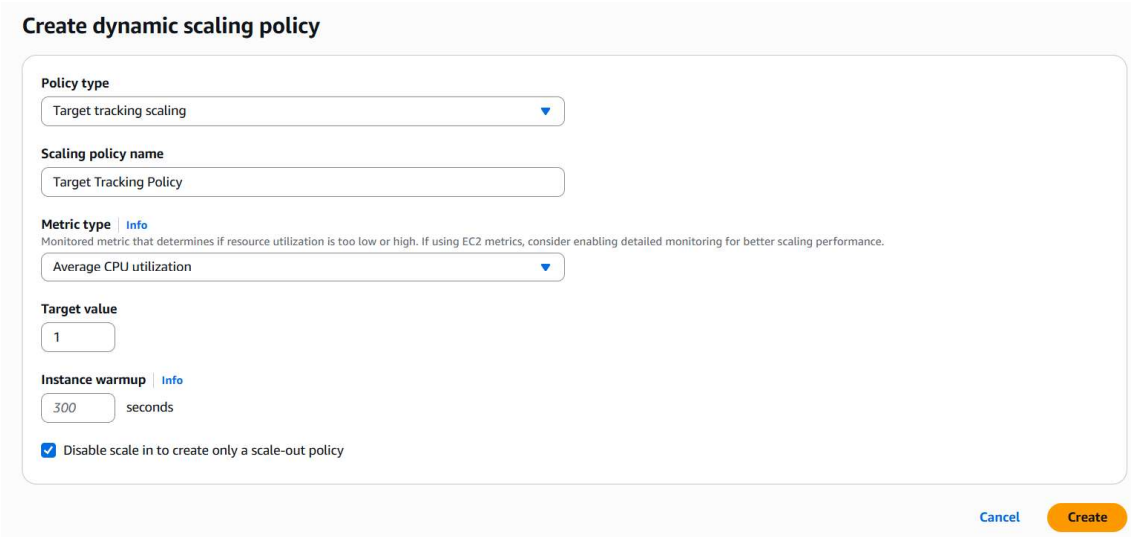
After we have to set some threshold to create an instance properly After utilizing the CPU of the instance.

Using the Dynamic Scaling Policies we can create the threshold for the ASG.

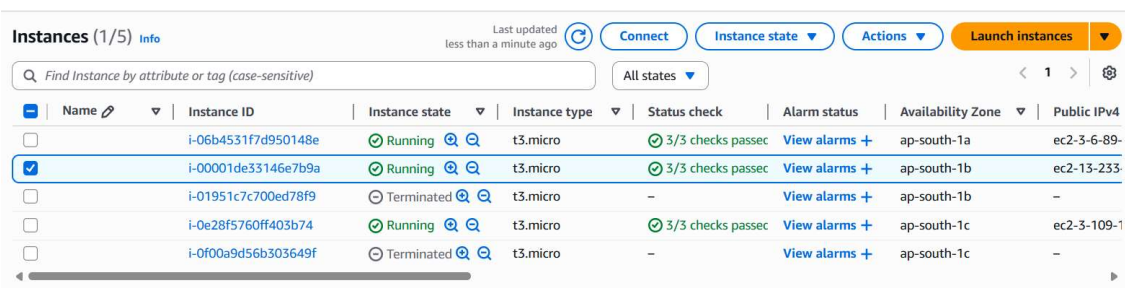


This is my Dynamic Policy with the 1% of CPU Utilization.

When the 1% Utilization is Hold Up For 5 mins It will Create One New Instance.



I have Give Fake Stress To Create The Instance.

```
[root@ip-172-31-6-141 ec2-user]# stress --cpu 1 --timeout 300
stress: info: [16117] dispatching hogs: 1 cpu, 0 io, 0 vm, 0 hdd
```

This is the created Instance After the Creation Dynamic Scaling Policy.